*Accounting Horizons*
Vol. 29, No. 3
2015
pp. 529–549

American Accounting Association
DOI: 10.2308/acch-51074

# Auditor Resignation and Risk Factors

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**SYNOPSIS:** Although auditor litigation risk is considered as a leading explanation for auditor resignations, audit risk and business risk might also trigger resignations. Auditor litigation risk is defined as the risk of the auditor being involved in a lawsuit, audit risk is defined as the risk that the auditor expresses an inappropriate audit opinion when the financial statements are materially misstated and, finally, business risk is defined as the risk associated with the client's survival and profitability. Because the three risk factors are not mutually exclusive, we examine their relevance and incremental importance using measures from the pre- and post-resignation periods. Using summary indices from the pre-resignation period, we find that all of the three *ex ante* risk indices are incrementally important for resignations, especially when the predecessor auditor is a Big 4 firm. Because the *ex ante* risk factors are prone to measurement errors and are less likely to capture the auditor's proprietary information about the client, we analyze data from the post-resignation period when the auditor's proprietary information is likely to become publicly known. We find that within a three-year period following an auditor's resignation, clients are more likely to (1) be involved in class-action lawsuits (*ex post* litigation risk), (2) have internal control problems (*ex post* audit risk), and (3) to be delisted from a national stock exchange (*ex post* business risk). Our research demonstrates that auditors consider all three risk factors, and not just litigation risk, in resignation decisions.

**Keywords:** auditor resignations; litigation risk; audit risk; business risk.

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We thank Arnold M. Wright (editor), two anonymous referees, Michel Magnan, Hakyin Lee, Steve Lilien, Steven Lustgarten, Liz Peltier-Wagner, Bill Ruland, Jangwon Suh, Yannan Shen, Ping Wang, Bo Zhang, Sha Zhao, and the participants and the discussant, Tim Louwers, at the 2008 AAA Annual Meeting in Los Angeles, CA for their helpful comments and suggestions. We thank Hangsoo Kyung for helping us with the data collection.

*Submitted: November 2012*
*Accepted: February 2015*
*Published Online: September 2015*
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## INTRODUCTION

Litigation risk, defined as the risk that the auditor is involved in a lawsuit, is often considered as a dominant explanation for auditor resignations.[1] However, the results from a recent study by Catanach, Irving, Williams, and Walker (2011) suggest that business risk, defined as the risk associated with the client's survival and profitability, might be another competing explanation for resignations. Further, audit risk, defined as the risk that the auditor expresses an inappropriate audit opinion when the financial statements are materially misstated, is a third plausible explanation for auditor resignations. Because the three risk factors are not mutually exclusive, it is ambiguous whether they are incrementally important as competing explanations for auditor resignations. Further, although prior studies claim that litigation risk is the dominant explanation, we are not aware of any study that evaluates the relative importance of the three risk factors by considering them simultaneously.

In this study, we examine the relevance and incremental importance of the three competing risk-based explanations for auditor resignations using information from the pre- and post-resignation periods. We construct *ex ante* summary indices for litigation, business, and audit risk using information one year prior to auditor resignations. We then evaluate the incremental importance of the three indices in precipitating auditor resignations. One limitation of relying on *ex ante* risk measures is that they are noisy estimates of auditors' proprietary information about clients' "hidden risks," because market participants cannot observe the auditors' private information.

A key innovation of our study is that we analyze client-related outcomes from the post-resignation period, which are less susceptible to measurement errors because the auditors' proprietary information about the clients' hidden risks is likely to be revealed over the post-resignation period. Drawing on the Bockus and Gigler (1998) analytical model of auditor resignations, we identify three distinct client-related adverse outcomes from the post-resignation period. First, auditors are frequently included as defendants in shareholder class-action lawsuits (Stice 1991; Sullivan 1992; Palmrose 1988). Therefore, we analyze the incidence of class-action lawsuits against clients from the post-resignation period as an *ex post* test of whether auditors resign to limit their litigation risk (*ex post* litigation risk proxy).[2] Second, we examine whether clients are more likely to be associated with internal control weaknesses in financial reporting over the post-resignation period (*ex post* audit risk proxy). Third, we also investigate whether clients are more likely to be delisted from a stock exchange following resignations (*ex post* business risk proxy).

In a recent study, Catanach et al. (2011) also examine whether resignations lead to more frequent delisting when the successor auditor is a Big 4 or non-Big 4 firm. Because their analysis is confined to resignations, it is hard to assess whether all auditor switches (dismissal and resignations) are associated with delisting or whether delisting is unique to resignations. We build on Catanach et al. (2011) by examining the incremental importance of the three risk-based explanations for resignations using client-related outcomes from the post-switching period based on a sample of auditor switches.

Additionally, we consider the size of the predecessor and successor auditor when evaluating the importance of the risk factors because larger auditors with greater exposure to legal liability are more sensitive to all types of risk than smaller auditors (e.g., Simunic and Stein 1996; Lambert

[1] For example, see, Landsman, Nelson, and Rountree (2009), Rama and Read (2006), Bedard and Johnstone (2004), Choi, Jeon, and Park (2004), Shu (2000), J. Krishnan and J. Krishnan (1997), and Simunic and Stein (1996).

[2] A more direct test would be to examine the incidence of auditor lawsuits. However, it is not possible to observe an auditor's exposure to future lawsuits once they resign from a risky engagement (a counter factual case). Because auditors are invariably included in class-action lawsuits, the incidence of class-action lawsuits against the client over the post-resignation period provides an estimate of the legal risk that the auditor would have encountered had they not resigned.

1994). In the Bockus and Gigler (1998) model, larger auditors with more resources are better at detecting hidden risks because they can devote more resources to detecting such risks than smaller auditors. Also, because of greater wealth at risk, larger auditors have greater exposure to legal liability from failing to detect hidden risks than do smaller auditors. Consequently, larger auditors have both the incentives and the ability to detect hidden risks and when they detect that risks are sufficiently high, they resign from the engagement. Thus, resignations by larger auditors signal that engagements have high litigation risk, high audit risk, and/or high business risk.

Our results are based on a comprehensive sample of auditor switches between 1999 and 2010 with 1,158 resignations and 4,988 dismissals. Drawing on prior studies (Catanach et al. 2011; DeFond and Jiambalvo 1991), we construct three *ex ante* risk indices using public information for the fiscal year preceding the auditor switch. Consistent with the audit risk definition (AS No. 8, PCAOB 2010), audit risk is high for firms managing earnings, for high growth firms, for smaller firms, for those with external financing needs, and for firms with internal control deficiencies. Business risk is high for firms with going concern modified reports and those with high bankruptcy or delisting risk (e.g., Johnstone and Bedard 2004). Finally, litigation risk is high for firms with low stock returns, those in high-tech industries (Shu 2000), and firms involved in lawsuits or restatements (Catanach et al. 2011; Palmrose and Scholz 2004).

The univariate results indicate that the three risk factors are all higher (economically and statistically) for the *Resignation* sample relative to the *Dismissal* sample. In our multivariate analysis, when we include the three risk measures in a logistic regression where the dependent variable is 1 for resignations and 0 for dismissals, we find that all three risk measures are positive and statistically significant. Our results suggest that audit, business, and litigation risks are incrementally important in explaining auditor resignations. Additionally, when we use Big 4 *Resignation* as a dependent variable, the three risk measures continue to be significant. The economic and statistical significance of *Litigation risk* is much higher for Big 4 resignations than non-Big 4 resignations, which suggests that litigation risk is the most important consideration for Big 4 resignations.

Our *ex post* adverse outcome measures are consistent with the *ex ante* risk measure results. We find that within a three-year period following a resignation, clients are more likely to (1) be involved in securities shareholder class-action lawsuits, (2) report weaknesses in internal control over financial reporting, and (3) be delisted from a national stock exchange because of bankruptcy or failure to meet the stock exchange listing requirements. Because we include *ex ante* risk indices and various control variables from the pre-resignation period in our regressions, resignation is likely to capture the auditors' private information. Thus, our results suggest that auditors have proprietary information about clients' litigation risk, business risk, and audit risk and that they resign from engagements when they privately deem such risks to be high. When we incorporate the size of the predecessor auditor, our *ex post* results are stronger when the predecessor auditor is a Big 4 than a non-Big 4 firm.

In a recent study, Elliott, Ghosh, and Peltier (2013) document that auditors charge more when engagements are risky. Our study highlights an alternative solution. When risk is sufficiently high, auditors choose to resign because the engagement is deemed unprofitable for reputational and financial reasons. While we concentrate on the size of the predecessor auditor, we extend our analyses by partitioning the sample into four groups based on the size of the predecessor and successor auditors. In most instances, our results indicate that the odds of future adverse outcomes are higher when a Big 4 firm resigns from engagements (relative to a non-Big 4 firm or dismissals) regardless of the size of the successor auditor.

We conduct numerous additional tests to examine the robustness of the results. First, because there might be fundamental differences between the resignation and dismissal subsamples, we construct a matched sample whereby each resignation observation is matched to a dismissal

observation based on Shu's (2000) litigation risk measure. Second, we estimate our regressions after including firm fixed effects. Finally, we exclude clients switching from Arthur Andersen in 2002, and all second-tier auditors like Grant Thornton and BDO Seidman from our sample. Our results are robust to these additional sensitivity analyses.

Our study contributes to the auditing literature by documenting that business, audit, and litigation risks are all important considerations in auditor resignation decisions. While prior studies tend to emphasize litigation risk as a dominant explanation for Big 4 resignations (e.g., Landsman et al. 2009; Shu 2000; Krishnan and Krishnan 1997; Simunic and Stein 1996), our results suggest that non-litigation risks also precipitate auditor resignations. Because much of the information about auditors' assessment of clients' risk is private, *ex post* results are likely to provide powerful insights into auditors' assessments of clients' hidden risks at the time of the resignation.

## RESIGNATIONS AND RISK FACTORS

Our study develops proxies for risk factors from two distinct time intervals: (1) the pre-resignation or the *ex ante* period, and (2) the post-resignation or the *ex post* period.

### *Ex Ante* Risk Factors

For the pre-resignation period, we develop summary indices (scores) for litigation, business, and audit risk using public data one year prior to the resignation. Because each of the risk factors has been individually examined in prior research, the task of developing individual ex ante risk indices is relatively straightforward.

#### *Litigation Risk Score*

We identify three sources of litigation risk: (1) decline in stock price (*Stock Return*), (2) risky industries (*High-tech*), and (3) lawsuit risk (*Lawsuit* and *Restatement)*. Prior studies find that the litigation risk increases for firms with declining stock price, for high-tech industries (Shu 2000), for clients involved in class-action lawsuits (Palmrose 1988), and for firms restating prior period financial statements (Palmrose and Scholz 2004). We classify the population of publicly traded firms in Compustat into five quintiles based on annual stock returns ending with the fiscal year-end one year prior to the switch. We compute annual stock returns using Compustat data items. *Stock Return* is equal to 1 when firms are in the lowest quintile of annual stock returns, and 0 otherwise. As in Shu (2000), *High-tech* is equal to 1 for firms in the technology industries, and 0 otherwise, where technology industries are defined based on the following three-digit standard industry classification (SIC) codes: 283, 357, 737, 873, and the following four-digit SIC codes: 3825–3839. Using lawsuit data from the 2013 edition of the Corporate Legal Files of Audit Analytics, we code *Lawsuit* as equal to 1 if a client is involved in a class-action lawsuit one year prior to a switch for issues related to accounting; financial reporting; tax; and the SEC's accounting, auditing, or enforcement releases (i.e., Audit Analytics category types 1, 2, 41, 43, 48, 54), and 0 otherwise. *Restatement* is equal to 1 when a firm restates its financial statement one year prior to a switch, and 0 otherwise. The *Litigation risk score* value, thus, lies between 0 and 4.

#### *Audit Risk Score*

We identify five key sources of audit risk: (1) *Discretionary Accruals,* (2) *Market-to-Book,* (3) *Assets,* (4) *External Financing,* and (5) *Internal Control*. Because low earnings quality, or aggressive earnings management, is typically associated with firms with weak internal controls (Doyle, Weili, and McVay 2007), audit risk is high for firms with low earnings quality. We classify the population of publicly traded firms in Compustat into five quintiles based on the absolute value

of discretionary accruals computed from the modified Jones (1991) model, which additionally controls for performance, as in Kothari, Leone, and Wasley (2005).[3] Firms in the highest quintile have *Discretionary Accruals* equal to 1, and 0 otherwise.

Growth firms face more challenges in establishing and enforcing internal controls than mature firms do (Lys and Watts 1994). Similarly, smaller firms with more resource constraints are more prone to internal control problems. We use the market-to-book ratio (market value of equity plus book value of debt to total assets) and total assets as proxies for growth and firm size, respectively. We classify the population of publicly traded firms into five quintiles based on the market-to-book ratio and total assets. *Market-to-Book* (*Assets*) is equal to 1 when firms are in the highest (lowest) market-to-book ratio (total assets) quintile, and 0 otherwise. Also, firms relying on external financing have more incentives to manipulate their financial statements and therefore have higher audit risk. As in Bradshaw, Richardson, and Sloan (2006), we classify the population of publicly traded firms into five quintiles based on external financing.[4] *External Financing* is equal to 1 when firms are in the highest financing quintile, and 0 otherwise. Audit risk is high for firms with ineffective internal control over financial reporting, as reflected by a material control weakness(es) (Caplan 1999). *Internal Control* is equal to 1 when internal control over financial reporting is considered ineffective.[5]

All five variables are measured one year prior to an auditor switch, and the *Audit risk score* value lies between 0 and 5.

### *Business Risk Score*

We identify two key sources of business risk: (1) Bankruptcy risk (*Going Concern* and *Z-Score,* and (2) other delisting risk (*Asset Return*). A client's business risk is high for firms that are expected to go bankrupt. *Going Concern* is equal to 1 when an auditor issues a going concern modified report for the fiscal year immediately preceding the switch, which is obtained from the Audit Opinion file of Audit Analytics (GOING_CONCERN). We also sort Compustat firms into quintiles based on the Altman Z-Score. The *Z-Score* is equal to 1 when a firm is in the lowest Altman Z-Score quintile, and 0 otherwise.[6] A client's business risk is also high when a firm is likely to get delisted because of violations of stock exchange rules or because it might be acquired by another firm. We also use poor performance as a proxy for delisting risk. We classify the population of publicly traded firms into five quintiles based on income before extraordinary items to total assets (OIBDP/TA). *Asset Return* is equal to 1 if a firm is in the lowest return on asset return quintile (OIBDP/TA), and 0 otherwise.

All of the variables are measured one year prior to an auditor switch. The *Business risk score* value lies between 0 and 3.

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[3] Specifically, we regress total accruals on change in sales (ΔSALE), fixed assets (PPENT), and operating income (OIBDP). All variables are deflated by the beginning of the year total assets (AT). Total accruals are computed as changes in current assets (ΔACT), less changes in current liabilities (ΔLCT), less changes in cash and short-term investments (ΔCHE), less depreciation and amortization (DP), plus changes in current debt (DLC). The Compustat variables names are in parentheses. The regression model is estimated annually for each industry, where industry is defined using a two-digit SIC code.

[4] In particular, external financing is defined as the sum of the change in common equity (ΔCEQ) and the change in preferred stock (ΔPSTK), and the change in total debt. Change in total debt is the sum of the change in long-term debt (Δ{DLTT + DD1 − DCVT}), and the change in convertible debt (ΔDCVT), and the change in notes payable (Δ{DLC − DD1}). The Compustat variables names are in parentheses and Δ represents the first difference.

[5] Internal control data are obtained for the SOX404 (internal control) and SOX302 (disclosure control) files in Audit Analytics. For the years prior to the mandated disclosures on internal controls, we obtain internal control information from the auditor change 8-K filings.

[6] The Altman Z-score is defined as 1.2 × ({ACT − LCT}/TA) + 1.4 × (RE/TA) + 3.3 × ({PI + XINT}/TA) + 0.6 × (PRCC_F/{CEQ/CSHO}) + 1.0 × (SALE/TA). The Compustat variables names are in parentheses.

### *Ex Post* Risk Factors

We draw on the Bockus and Gigler (1998) model to derive adverse outcomes that follow resignations that are consistent with the risk factors (litigation, business, and audit risk). Because any risk index based on public information from the pre-resignation period is unlikely to fully capture the auditors' private assessments of the clients' hidden risks, we underscore the analyses of adverse outcomes from the post-resignation period to provide superior insights into auditors' assessments of clients' hidden risks.

#### *Bockus and Gigler Model*

In the Bockus and Gigler (1998) analytical model, the client firm privately knows whether it presents a significant hidden risk to the incumbent auditor, where hidden risk is defined as any *adverse* information that the client prefers to hide from the auditor. Unless incumbent auditors detect the hidden risk (e.g., management fraud), they have substantial exposure to added legal liability. The client does not want the incumbent auditor to detect the hidden risk because they must pay a penalty when the auditor detects such hidden risks. In contrast, the incumbent auditor has incentives to detect hidden risks posed by clients. Failing to do so, the auditor bears added litigation/reputational costs.

Large incumbent auditors are better at detecting whether a client presents hidden risks under two types of situations. First, incumbent auditors have an informational advantage over successor auditors in assessing hidden risk because they have access to private information from clients and they have proprietary information from their own audits. Second, auditors with more resources (or "larger" auditors) are better at detecting hidden risks than those with less resources (or "smaller" auditors). Upon detecting that a client poses a significant hidden risk, an incumbent auditor prefers to resign from the engagement.[7] Thus, a fundamental prediction of the Bockus and Gigler (1998, 204) model is:

> Since auditor resignations occur when the incumbent auditor believes it is relatively likely that the client has a hidden risk, we would expect that the firms whose auditors resign have a higher incidence of *adverse outcomes* than other firms. (emphasis added)

Based on their model, we identify the following three adverse outcomes that follow resignations.

#### *Post-Resignation Lawsuits: Litigation Risk*

We rely on the incidence of securities shareholder class-action lawsuits against clients related to financial reporting matters over the post-resignation period to judge the predecessor auditors' exposure to litigation risk had they not resigned from the engagement.[8] If auditors resign from an

[7] In their model an auditor resigns, and does not increase fees, to avoid an adverse selection problem. If an incumbent auditor with high legal liability were to increase fees, then the client who privately knows about the risk and observes competing bids would switch to another auditor willing to charge less. Consequently, the incumbent auditor is only left with excessively risky clients who are unable to find a successor auditor willing to charge less. Thus, increasing fees when client risks are sufficiently high hurts the incumbent auditor because they are left with only risky clients, which is why an incumbent auditor resigns from risky engagements in the Bockus and Gigler model.

[8] A securities class action is a lawsuit brought on behalf of a group of investors who have suffered a financial loss in a stock, bond, or investment fund. A loss may result from fraudulent stock manipulation, material false statements, or the restatement of previously issued financial statements. Claims usually arise under Rule 10b-5, where the fraud occurred in purchases made on a stock exchange, or under Section 11 of the Securities Act of 1933 where the securities purchased are traceable to a materially false and misleading registration statement (prospectus) filed with the SEC (Johnson, Kasznik, and Nelson 2001; Francis et al. 1994).

engagement because they question management integrity or they suspect fraud or fraudulent transactions, then investors are expected to seek damages for their losses once these infractions are revealed to the market in the future (Bonner, Palmrose, and Young 1998; Francis, Philbrick, and Schipper 1994).

Although the frequency of lawsuits against auditors is another direct measure to assess auditors' litigation risk, our contention is that the incidence of lawsuits against the client is a superior measure to judge the predecessor auditor's exposure to future litigation risk. For instance, investors may not include a smaller successor auditor in their class-action lawsuits because smaller auditors may not have the financial strength to settle claims or pay damages. However, had the larger predecessor auditor not resigned, it is more likely that the predecessor auditor would have been included as a defendant in the lawsuit because of their deep pockets. Therefore, securities class-action lawsuits against the client or management are better indicators of the predecessor auditor's future litigation risk than auditor lawsuits filed against the successor auditor following a resignation.

### *Post-Resignation Internal Control Problems: Audit Risk*

Depending on whether the underlying action causing the misstatement is intentional or unintentional, SAS No. 82 distinguishes between fraud and errors. Fraud includes both the deliberate misstatement of financial reporting and misappropriation of assets by the client firm. Management with a strong intention of perpetrating fraud would prefer weaker internal control over financial reporting so that fraud is not detected. Moreover, regardless of the internal control strengths, they would want to override those controls to ensure that the fraud remains hidden (Caplan 1999). Therefore, larger auditors are more likely to resign from risky engagements when they suspect issues with internal controls or that management might override internal controls regardless of the strength of the control system.[9]

### *Post-Resignation Delisting: Business Risk*

An auditor might also resign from an engagement when faced with high business or financial risk (Johnstone and Bedard 2004; Morgan and Stocken 1998). Business risk increases the auditors' potential litigation costs regardless of whether there is an audit failure. Also, when business risk is high because client firms' continued survival and well-being are in question, auditors deem such engagements as unprofitable (Bell, Landsman, and Shackelford 2001). Therefore, auditors are expected to resign from engagements with high business risk.

A direct proxy for a client's business risk is that they cease to exist as a publicly traded firm, which automatically terminates the need for an auditor-client relationship from a mandated perspective. Because of the exposure to potential litigation costs when a client is delisted from a stock exchange (e.g., bankruptcy or liquidation), and since these costs are higher for larger auditors, we expect larger auditors to resign more frequently when business risk is high.

[9] Auditors might respond to higher levels of control risk by increasing the level of audit effort needed to maintain an acceptable level of audit risk, which leads to higher audit fees (Hogan and Wilkins 2008). We propose an alternative solution that, when control risk is sufficiently high, auditors choose to resign from risky engagements because the amount of effort required to decrease the level of audit risk to an acceptable level might deem the engagement as unprofitable.

## METHODOLOGY

### *Ex Ante* Risk and Resignations

Our objective is to analyze whether the three *ex ante* risk measures are incrementally (jointly and incrementally) important, and their relative importance, in auditor resignation decisions. Therefore, we estimate the following logistic regression:

$$\begin{aligned} Resignation = \beta_0 + \beta_1 Litigation\ risk\ score + \beta_2 Audit\ risk\ score \\ + \beta_3 Business\ risk\ score + \varepsilon \end{aligned} \tag{1}$$

where the dependent variable is *Resignation,* which equals 1 when the predecessor auditor resigns from an engagement, and 0 otherwise. The main independent variables are *Litigation risk score, Audit risk score,* and *Business risk score* (as defined previously). If auditors consider all types of risk factors in resignation decisions, then $\beta_1$, $\beta_2$, and $\beta_3$ are expected to be positive. We also estimate Equation (1) by replacing *Resignation* with $Resignation_{B4}$, which equals 1 when a Big 4 auditor resigns.

### *Ex Post* Risk: Adverse Outcomes and Resignations

In the second part of our analyses, we directly examine adverse outcomes following auditor resignations. We use the following parsimonious regression model to test the relationship between auditor resignations and future adverse outcomes while controlling for the pre-switch information environment:

$$\begin{aligned} Adverse\ outcomes_{i=j} = \beta_0 + \beta_1 Resignation_0 + \beta_2 Acquisition_{-1} + \beta_3 Leverage_{-1} + \beta_4 Size_{-1} \\ + \beta_5 Growth_{-1} + \beta_6 Market\text{-}to\text{-}Book_{-1} + \beta_7 Return\text{-}on\text{-}Assets_{-1} \\ + \beta_8 Litigation\ risk\ score_{-1} + \beta_9 Audit\ risk\ score_{-1} \\ + \beta_{10} Business\ risk\ score_{-1} + \beta_{11} Adverse\ outcomes_k \\ + \beta_{12} Adverse\ outcomes_l + \sum Industry/Year\ fixed\ effects + \varepsilon \end{aligned} \tag{2}$$

We measure *Adverse outcomes* using three distinct constructs ($i = j$, $k$, or $l$): (1) Class-action lawsuits (*Lawsuits*), which equals 1 when a firm is a defendant in a class-action lawsuit for violation of securities laws anytime within three years subsequent to a switch, and 0 otherwise (litigation risk proxy), (2) *Internal Control* equals 1 when a firm reports ineffective internal control over financial reporting anytime within three years subsequent to a switch, and 0 otherwise (audit risk proxy), and (3) *Delisting* equals 1 when a firm is delisted from a stock exchange, for reasons other than merger/acquisition, within three years subsequent to a switch, and 0 otherwise (business risk proxy).[10]

Consistent with prior studies, we include several control variables that are associated with adverse outcomes but are also likely to be associated with auditor switches (i.e., resignations or dismissals).[11] We control for *Acquisition,* which equals 1 when a firm engages in a merger or acquisition, and 0 otherwise because client firms are more likely to have financial reporting problems following an acquisition; we include *Leverage* (ratio of the sum of long-term debt and short-term debt to total assets) because highly leveraged firms have incentives to misstate their

[10] *Lawsuit* is equal to 1 if a client is involved in a class-action lawsuit for issues related to accounting, financial reporting, tax, and SEC accounting, auditing, or enforcement releases (i.e., Audit Analytics category types 1, 2, 41, 43, 48, 54), and 0 otherwise. *Internal Control* is equal to 1 when internal control is deemed ineffective, and 0 otherwise. *Delisting* is equal to 1 when the CRSP delisting codes are 170, 400–490, 535–587, and 589–591, and 0 otherwise.

[11] See, for example, Abbott, Parker, and Peters (2004), Kinney, Palmrose, and Scholz (2004); Khurana and Raman (2004), Palmrose and Scholz (2004), Heninger (2001), Johnson et al. (2001), Shu (2000), Lys and Watts (1994), Carcello and Palmrose (1994), Francis et al. (1994), and DeFond and Jiambalvo (1991).

financial reports; we control for *Size* (logarithm of total assets), *Growth* (percentage change in revenues between the current and the prior year), and *Market-to-Book* (market value of equity plus the book value of preferred stock and debt to the book value of total assets) because prior research finds that smaller and rapidly growing firms are more likely to develop financial reporting problems; we control for *Return-on-Assets* (income before extraordinary items to total assets) because poorly performing firms are more likely to encounter adverse outcomes. All control variables are measured for the fiscal year prior to the auditor switch to control for the information environment that preceded the switch.

We additionally include the three *ex ante* risk factors that are correlated with auditor resignations. Because the *ex ante* risk measures might also predict future adverse outcomes, it is critical to control for these factors before drawing any inferences about the information content of auditor resignation decisions. Finally, because the future adverse outcomes ($j$, $k$, $l$) may be correlated with each other, we control for the other two outcomes ($i = k$ or $l$) when we analyze a particular adverse outcome ($i = j$).[12]

The main variable *Resignation* equals 1 when an auditor resigns, which is the year of the switch. Because we control for the pre-resignation information and the *ex ante* risk factors, *Resignation* captures auditors' private information regarding clients' hidden risks. To test whether resignations by larger (smaller) auditors are associated with increased risks, we replace *Resignation* with $Resignation_{B4}$ ($Resignation_{NB4}$) to analyze the effects of Big 4 (non-Big 4) resignations on future adverse outcomes. We also include four indicator variables that incorporate the size of the predecessor and successor auditors ($Resignation_{B4toB4}$, $Resignation_{NB4toNB4}$, $Resignation_{B4toNB4}$, and $Resignation_{NB4toB4}$).

## DATA

### Sample Selection and Descriptive Statistics

We collect data on auditor switches (resignations and dismissals) from the 2013 Audit Analytics (AA) database covering all SEC registrants reporting changes in the external auditor between 1999 and 2010. From this initial sample, which contains a total of 19,075 switches, we identify whether the auditor resigned or was dismissed from the auditor change 8-K filings. This sample selection procedure results in 14,254 dismissals and 4,821 resignations. We obtain accounting data from Compustat. Because many firms listed in AA are not included in Compustat, our sample reduces to 8,234 observations (6,474 dismissals and 1,760 resignations) when we merge the Compustat and Audit Analytics databases.[13] However, the sample size declines to 6,680 observations because we require nonmissing values for the control variables used in our regression analysis. Additionally, to remove the effect of extreme observations, we delete the top and bottom 1 percent of the financial ratios including *Leverage, Growth, Market-to-Book,* and *Return-on-Assets*.[14] Our final sample includes 6,146 auditor changes, of which 1,158 are resignations and the remaining 4,988 cases are auditor dismissals.

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[12] For instance, when we use *Lawsuit,* which serves as one of the three proxies for future adverse outcomes, as the dependent variable, we include the other two proxies for future adverse outcomes (i.e., *Internal Control* and *Delisting*) as independent variables.

[13] While the AA database includes information mostly about large publicly traded companies, it also includes information about management investment companies (Trust/Fund), small business issuers, privately held companies, foreign private issuers, and wholly owned subsidiaries that are not covered in Compustat. Therefore, we lose about 10,841 observations when we match AA with Compustat.

[14] Our results are robust to alternative treatments of outliers including winsorizing the top and bottom 1 percent of the influential observations or using other cut-off points, e.g., winsorizing/deleting the top and bottom 2, 3, or 5 percent of observations.

**TABLE 1**

**Descriptive Statistics**

| | *Dismissal* | | *Resignation* | | Differences | |
|---|---|---|---|---|---|---|
| | **Mean** | **Median** | **Mean** | **Median** | **Mean** | **Median** |
| *Assets* (million dollars) | 1,161 | 113 | 484 | 32 | 677*** | 81*** |
| *Acquisition* | 0.1263 | 0.0000 | 0.1079 | 0.0000 | 0.0184*** | 0.0000 |
| *Leverage* | 0.3758 | 0.1909 | 0.4749 | 0.1891 | −0.0991*** | 0.0018 |
| *Growth* | 0.2912 | 0.0524 | 0.2829 | 0.0330 | 0.0083*** | 0.0194*** |
| *Market-to-Book* | 3.0078 | 1.0598 | 4.4024 | 1.2257 | −1.3947*** | −0.1659*** |
| *Return-on-Assets* | −0.3176 | 0.0024 | −0.6302 | −0.1022 | 0.3125*** | 0.1047*** |
| *Big 4* | 0.6195 | 1 | 0.4784 | 0 | 0.1411 | 1*** |
| Observations | 4,988 | | 1,158 | | | |

*** Indicates significance at the 1 percent level.
The tests of the mean (median) differences between *Dismissal* and *Resignation* are based on t-values (Wilcoxon two-sample Z-tests).

Variable Definitions:
*Assets* = book value of total assets (in million dollars);
*Acquisition* = an indicator variable that equals 1 when a firm engages in a merger or acquisition, and 0 otherwise;
*Leverage* = ratio of the sum of long-term debt and short-term debt to total assets;
*Growth* = percentage change in revenues between the current and the prior year;
*Market-to-Book* = ratio of the sum of the market value of equity plus the book values of preferred stock and debt divided by the book value of total assets;
*Return-on-Assets* = income before extraordinary items to total assets; and
*Big 4* = an indicator variable that equals 1 if the auditor is Big 4, and 0 otherwise.

Data on the effectiveness of internal control over financial reporting are collected from the Internal Control Effectiveness (SOX404) file of Audit Analytics. There are a total of 11,196 observations between 2001 and 2013 with ineffective internal control over financial reporting, of which 523 firm-years report ineffective internal controls within three years following an auditor switch; 370 (153) cases are within three years of a dismissal (resignation). We use the 2013 edition of the Corporate Legal Files of Audit Analytics to obtain names of clients that are defendants in class-action lawsuits for issues related to accounting, financial reporting, tax, and SEC enforcements. There are a total of 3,488 unique lawsuits filed between 2000 and 2013, of which 272 lawsuits match with our sample and are filed within three years following an auditor switch; 220 (52) cases are filed within three years of a dismissal (resignation). Finally, we use the 2013 CRSP stock files and find a total of 2,884 delisting cases between 2000 and 2013, of which 512 cases are filed within three years following an auditor switch; 379 (133) cases are within three years of a dismissal (resignation).

Table 1 reports the descriptive statistics (mean and median) for auditor dismissal (*Dismissal*) and auditor resignations (*Resignation*) samples. We also report the mean and median differences between the two subsamples. Compared to the *Dismissal* sample, a typical firm is considerably smaller for the *Resignation* sample. The average merger and acquisition activity for the *Dismissal* sample is statistically significantly higher than the *Resignation* sample (0.13 versus 0.11). The mean *Leverage* is higher for the *Resignation* sample. Although resignation firms have lower mean and median growth in revenues (*Growth*), the mean and median *Market-to-Book* ratio (proxy for future growth opportunities) is much higher. A typical firm in the *Resignation* sample is much less

**TABLE 2**

***Ex Ante* Risk Metrics**
**Differences in *Ex Ante* Risk Scores between Resignation and Dismissal**

| | *Dismissal* | | *Resignation* | | | |
|---|---|---|---|---|---|---|
| **Variable** | **n** | **Mean** | **n** | **Mean** | **Difference** | **t-value** |
| *Litigation risk score* | 4,988 | 0.4389 | 1,158 | 0.5328 | −0.0940 | 3.23*** |
| *Audit risk score* | 4,988 | 0.8771 | 1,158 | 1.1546 | −0.2775 | 3.10** |
| *Business risk score* | 4,988 | 0.7871 | 1,158 | 1.2504 | −0.4633 | 3.24*** |

***, ** Indicate significance at the 1 percent and 5 percent levels, respectively.
All variables are measured for the fiscal year immediately preceding the auditor switch.
*Litigation risk score* is based on four variables: (1) We classify Compustat firms into five quintiles based on annual stock returns prior to the switch and *Stock Return* = 1 when firms are in the lowest quintile of annual stock returns, and 0 otherwise; (2) *High-tech* = 1 for firms in the technology industries, and 0 otherwise, where technology industries are defined based on the following three-digit SIC codes: 283, 357, 737, 873, and the following four-digit SIC codes: 3825–3839; (3) *Lawsuit* = 1 if a client is involved in a class-action lawsuit one year prior to a switch for issues related to accounting, financial reporting, tax, and SEC enforcements (i.e., category types 1, 2, 41, 43, 48, 54), and 0 otherwise; and (4) *Restatement* = 1 when a firm restates its prior period financial statements, and 0 otherwise. *Litigation risk* value lies between 0 and 4.
*Audit risk score* is based on five variables: (1) *Discretionary Accruals* = 1 for firms in the top quintile, and 0 otherwise, where quintiles are based on absolute discretionary accruals from the modified Jones' model; (2) *Market-to-Book* ratio = 1 for firms in the top quintile, and 0 otherwise, where *Market-to-Book* is the ratio of the market value of equity plus the book value of debt to the book value of total assets; (3) *Assets* = 1 for firms in the lowest quintile, and 0 otherwise, where *Assets* is the book value of total assets; (4) *External Financing* = 1 for firms in the top external financing quintile, and 0 otherwise, where *External Financing* is defined as the sum of the change in common equity (ΔCEQ), and the change in preferred stock (ΔPSTK), and the change in total debt. Change in total debt is the sum of the change in long-term debt (Δ{DLTT + DD1 − DCVT}), and the change in convertible debt (ΔDCVT), and the change in notes payable (Δ{DLC − DD1}); and (5) *Internal Control* = 1 when internal control is ineffective. *Audit risk score* lies between 0 and 5.
*Business risk score* is based on three variables: (1) *Going Concern* = 1 when an auditor issues a going concern modified report (GOING_CONCERN), and 0 otherwise; (2) *Z-Score* = 1 if a firm is in the lowest Altman Z-score quintile, and 0 otherwise, where the Altman Z-score is defined as 1.2 × ({ACT − LCT}/TA) + 1.4 × (RE/TA) + 3.3 × ({PI + XINT}/TA) + 0.6×(PRCC_F/{CEQ/CSHO}) + 1.0 × (SALE/TA); and (3) *Asset Return* = 1 if a firm is in the lowest return on asset return quintile (OIBDP/TA), and 0 otherwise. *Business risk score* lies between 0 and 3.

profitable as measured using *Return-on-Assets*. Finally, the proportion of larger (*Big 4*) auditors is higher for the *Dismissal* sample.

## EMPIRICAL RESULTS

### *Ex Ante* Risk Measures and Resignations

In Table 2, we report the mean values for the three *ex ante* risk factors for the resignation and dismissal subsamples. Our results indicate that all three *ex ante* risk factor scores are statistically significantly larger for the *Resignation* sample compared to the *Dismissal* sample, which is consistent with auditors resigning when all three risks are deemed high. The mean *Litigation risk* is 0.44 for the *Dismissal* sample, but the corresponding number for the *Resignation* sample is 0.53, which suggests that litigation risk is higher for the *Resignation* sample by about 20 percent. The difference in *Litigation risk* between the two subsamples is statistically significant (p-value $<$ 0.01). Similarly, *Audit risk* is 0.87 for the *Dismissal* sample, but the corresponding number for the *Resignation* sample is 1.15, which suggests that audit risk is higher for the *Resignation* sample by about 32 percent. The difference in *Audit risk* between the two subsamples is statistically significant (p-value $<$ 0.05). Finally, *Business risk* is 0.79 for the *Dismissal* sample, but the corresponding

**TABLE 3**

***Ex Ante* Risk Indices and Auditor Resignations**

| | *Resignation* | | *Resignation*$_{B4}$ |
|---|---|---|---|
| **Dependent Variable** | **Model 1** | **Model 2** | **Model 3** |
| Intercept | −2.1335 | −2.7586 | −3.4656 |
| | (88.93)*** | (29.57)*** | (22.09)*** |
| *Litigation risk score* | 0.1076 | 0.2030 | 0.3903 |
| | (3.67)** | (6.49)*** | (19.18)*** |
| *Audit risk score* | 0.0861 | 0.1667 | 0.1498 |
| | (4.80)** | (8.87)*** | (5.31)** |
| *Business risk score* | 0.329 | 0.4036 | 0.3231 |
| | (95.99)*** | (46.67)*** | (22.57)*** |
| *Litigation risk*$_{Shu}$ | — | −0.0382 | 0.052 |
| | | (0.64) | (0.86) |
| Observations | 6,146 | 3,329 | 3,329 |
| Adjusted $R^2$ (%) | 0.1339 | 0.1947 | 0.2012 |
| Likelihood Ratio | 532.63*** | 396.18*** | 345.10*** |
| Industry fixed effects | Included | Included | Included |
| Year fixed effects | Included | Included | Included |

***, ** Indicate significance at the 1 percent and 5 percent levels, respectively.

Variable Definitions:
*Resignation* = an indicator variable that equals 1 when an auditor resigns, and 0 otherwise;
*Resignation*$_{B4}$ = an indicator variable that equals 1 when a Big 4 predecessor auditor resigns from an engagement, and 0 otherwise;
*Litigation risk score* = an *ex ante* metric for litigation risk based on the four variables defined in Table 2;
*Audit risk score* = an *ex ante* metric for audit risk based on the five variables defined in Table 2;
*Business risk score* = an *ex ante* metric for business risk based on the three variables defined in Table 2; and
*Litigation risk*$_{Shu}$ = litigation risk computed using the Shu (2000) model.

number for the *Resignation* sample is 1.25, which suggests that business risk is higher for the *Resignation* sample by about 58 percent (p-value $< 0.01$).

In Table 3 we report the logistic regression results that examine whether the three *ex ante* risk indices are incrementally important in explaining resignation. In Model 1, we estimate a logistic regression of *Resignation* on *Litigation risk score, Audit risk score,* and *Business risk score*. We find that *Litigation risk score* (0.1076, $\chi^2 = 3.67$), *Audit risk score* (0.0861, $\chi^2 = 4.80$), and *Business risk score* (0.3290, $\chi^2 = 95.99$) are all positive and significant at the 5 percent level. Our results suggest that all three *ex ante* risk factors are incrementally important in explaining auditor resignations. Thus, the results suggest that auditors consider all risk factors, and not just litigation risk, in their resignation decisions.

In Model 2, when we additionally include Shu's (2000) litigation risk measure (*Litigation risk*$_{Shu}$), the number of observations reduces to 3,329 because of missing values in estimating Shu's model. The three *ex ante* risk indices remain highly significant, as in Model 1. However, the coefficient on *Litigation risk*$_{Shu}$ is not significant, which may be because of the high correlation between *Litigation risk*$_{Shu}$ and *Litigation risk* ($\rho = 0.26$). When we drop *Litigation risk* from Model 2, in untabulated results, we find that the coefficient on *Litigation risk*$_{Shu}$ is positive and significant (p-value $< 0.01$). In Model 3, we use *Resignation*$_{B4}$ as the dependent variable. All three risk indices remain highly significant, as in the previous models. The coefficient on *Litigation risk score*

increases from 0.2030 in Model 2 to 0.3903 in Model 3, which is also the largest coefficient among the three risk factors. These results indicate that litigation risk is the most important risk factor in Big 4 resignations.

Overall our results, based on publicly available information from the pre-switching period, indicate that all three risk factors (litigation, business, and audit risk) play an important role in precipitating auditor resignations. Litigation risk is the most important criterion in determining Big 4 resignations, followed by business risk and then by audit risk.

## *Ex Post* Risk and Resignations

### *Post-Resignation Class-Action Lawsuits*

Table 4 presents the results on the relationship between auditor resignations and class-action lawsuits from the post-resignation period, which is our proxy for the predecessor auditors' exposure to litigation had they not resigned from an engagement. The univariate results from Panel A of Table 4 indicate that the frequency of future class-action lawsuits is 5.90 percent for *Resignation* and 5.26 percent for *Dismissal*. This difference is even larger (around 3.2 percent and significant) when a Big 4 auditor resigns from an engagement. Thus, the likelihood of a future lawsuit when a Big 4 auditor resigns from an engagement is about 61 percent higher than when an auditor is dismissed (3.2/5.26 = 61 percent). The frequency of future lawsuits is also significantly higher when a non-Big 4 auditor resigns from an engagement.

In Table 4, Panel B we estimate Equation (2) using a logistic regression where the bivariate dependent variable *Lawsuit* equals 1 when a client firm is a defendant in a class-action lawsuit anytime within three years subsequent to an auditor switch, and 0 otherwise. In Model (1), the coefficient on *Resignation* ($\beta_1$) is positive and significant (0.1744, $\chi^2 = 4.11$; p-value $< 0.05$). In Model 2, we replace *Resignation* with $Resignation_{B4}$ and $Resignation_{NB4}$, the coefficients on $Resignation_{B4}$ ($\beta_1$) is positive and weakly significant (0.1803, $\chi^2 = 3.03$; p-value $< 0.10$), while that on $Resignation_{NB4}$ ($\beta_2$) is insignificant (0.1645, $\chi^2 = 1.61$; p-value $> 0.10$). In Model 3, when we include the four indicator variables that incorporate the size of the predecessor and successor auditors, none of the four resignation indicator variables is statistically significant at the 5 percent level.

Because our regression model controls for the pre-switching information environment (i.e., *ex ante* measures of litigation risk, audit risk, and business risk) and any delisting or internal control problems in the post-resignation period, *Resignation* captures auditors' proprietary information about the clients' hidden expected litigation risks. Our results suggest that there is little information content in resignation decisions with regard to future litigation risk because much of the pre-switching information environment, including *Audit risk score* (*ex ante* measure of audit risk) and *ex post* measures of audit risk and business risk (*Internal Control* and *Delisting*), are highly effective in predicting future litigation risk. Thus, even though an auditor might resign from engagements with excessively high litigation risk, much of this information is captured based on public information and there is no new information about future litigation risk based on resignation decisions.

### *Post-Resignation Internal Control Problems*

Table 5 presents the results of the relationship between auditor resignations and clients having ineffective internal control over financial reporting for the post-resignation period, our proxy for the predecessor auditors' exposure to future audit risk had they not resigned from an engagement. The univariate results from Panel A of Table 5 indicate that the frequency of future class-action lawsuits is 45.26 percent for the *Resignation* sample and 29.64 percent for *Dismissal* sample. The difference in the frequency of future internal control inefficiencies between the two samples (15.62 percent) is

## TABLE 4

### Auditor Resignations and Future Class-Action Lawsuits

### Panel A: Frequency of Future Lawsuits

| | *Resignation* | | *Dismissal* | | Difference | |
|---|---|---|---|---|---|---|
| | Obs. | Mean | Obs. | Mean | Mean | t-value |
| *Resignation* | 1,034 | 0.0590 | 4,410 | 0.0526 | 0.0064 | (0.79) |
| *Resignation*$_{B4}$ | 494 | 0.0850 | 4,410 | 0.0526 | 0.0324 | (2.49)** |
| *Resignation*$_{NB4}$ | 540 | 0.0352 | 4,410 | 0.0526 | 0.0174 | (2.02)** |

### Panel B: Logistic Regressions

| | Model 1 | | Model 2 | | Model 3 | |
|---|---|---|---|---|---|---|
| Intercept | −2.8803 | (163.46)*** | −2.8773 | (160.67)*** | −2.8703 | (159.89)*** |
| *Resignation* | 0.1744 | (4.11)** | | | | |
| *Resignation*$_{B4}$ | | | 0.1803 | (3.03)* | | |
| *Resignation*$_{NB4}$ | | | 0.1645 | (1.61) | | |
| *Resignation*$_{B4toB4}$ | | | | | 0.3131 | (3.82)* |
| *Resignation*$_{B4toNB4}$ | | | | | 0.1062 | (0.70) |
| *Resignation*$_{NB4toB4}$ | | | | | 0.0747 | (0.03) |
| *Resignation*$_{NB4toNB4}$ | | | | | 0.1698 | (1.56) |
| Control Variables | | | | | | |
| *Acquisition* | −0.1364 | (2.22) | −0.1363 | (2.22) | −0.1344 | (2.16) |
| *Leverage* | −0.0233 | (0.04) | −0.0225 | (0.04) | −0.0253 | (0.05) |
| *Size* | 0.2117 | (113.50)*** | 0.2113 | (110.32)*** | 0.2099 | (107.71)*** |
| *Growth* | 0.1245 | (24.09)*** | 0.1244 | (24.05)*** | 0.1239 | (23.87)*** |
| *Market-to-Book* | 0.0087 | (7.37)*** | 0.0087 | (7.38)*** | 0.0086 | (7.32)*** |
| *Return-on-Assets* | 0.1986 | (2.74)* | 0.1983 | (2.73)* | 0.2000 | (2.77)* |
| *Litigation risk score* | 0.0375 | (0.56) | 0.0369 | (0.54) | 0.0377 | (0.56) |
| *Audit risk score* | 0.1257 | (9.32)*** | 0.1257 | (9.31)*** | 0.1277 | (9.59)*** |
| *Business risk score* | −0.0191 | (0.12) | −0.0194 | (0.13) | −0.0178 | (0.11) |
| *Internal Control* | 0.4589 | (43.55)*** | 0.4587 | (43.50)*** | 0.4591 | (43.57)*** |
| *Delisting* | 0.3918 | (14.39)*** | 0.3919 | (14.39)*** | 0.3931 | (14.45)*** |
| Observations | 5,444 | | 5,444 | | 5,444 | |
| Adjusted $R^2$ (%) | 0.2288 | | 0.2288 | | 0.2294 | |
| Likelihood Ratio | 444.20*** | | 444.21*** | | 445.38*** | |
| Industry/Year fixed effects | Included | | Included | | Included | |

***, **, * Indicate significance at the 1 percent, 5 percent, and 10 percent levels, respectively.
The dependent variable *Lawsuit* is equal to 1 when a firm is involved in a lawsuit within three years subsequent to an auditor switch. Subscript *B4* (*NB4*) denotes a Big 4 (non-Big 4) predecessor auditor. Subscript *B4toB4* (*NB4toNB4*) denotes a Big 4 (non-Big 4) predecessor and successor auditors. Subscript *B4toNB4* (*NB4toB4*) denotes a Big 4 (non-Big 4) predecessor auditor and a non-Big 4 (Big 4) successor auditor.

Variable Definitions:
*Resignation* = an indicator variable that equals 1 when an auditor resigns, and 0 otherwise;
*Acquisition* = an indicator variable that equals 1 for merger or acquisition, and 0 otherwise;
*Leverage* = ratio of the sum of long-term debt and short-term debt to total assets;
*Size* = logarithm of the book value of total assets;
*Growth* = percentage change in revenues between the current and the prior year;
*Market-to-Book* = ratio of the sum of the market value of equity plus the book values of preferred stock and debt divided by the book value of total assets;

*(continued on next page)*

**TABLE 4 (continued)**

*Return-on-Assets* = income before extraordinary items to total assets;
*Litigation risk score* = an *ex ante* metric for litigation risk based on the four variables defined in Table 2;
*Audit risk score* = an *ex ante* metric for audit risk based on the five variables defined in Table 2;
*Business risk score* = an *ex ante* metric for business risk based on the three variables defined in Table 2;
*Internal Control* = an indicator variable that equals 1 when a firm reports a problem in internal control over financial reporting within three years subsequent to an auditor switch; and
*Delisting* = an indicator variable that equals 1 when a firm is delisted from a national stock exchange within three years subsequent to an auditor switch.

highly statistically significant (p-value $< 0.01$). Thus, the likelihood of a client developing future problems with internal control over financial reporting is about 54 percent higher when an auditor resigns from an engagement compared to when an auditor is dismissed. This difference is very similar when a Big 4 (17.12 percent) or a non-Big 4 auditor (14.25 percent) resigns from an engagement.

In Table 5, Panel B we estimate Equation (2) using a logistic regression where the bivariate dependent variable *Internal Control* is equal to 1 when a firm reports ineffective internal control over financial reporting anytime within three years subsequent to an auditor switch, and 0 otherwise. In Model 1, the coefficient on *Resignation* ($\beta_1$) is positive and significant (0.2992, $\chi^2 = 39.04$; p-value $< 0.01$). In Model 2, we replace *Resignation* with $Resignation_{B4}$ and $Resignation_{NB4}$, the coefficients on $Resignation_{B4}$ ($\beta_1$) and $Resignation_{NB4}$ ($\beta_2$) are both positive and significant (0.3188, $\chi^2 = 24.07$ and 0.2811, $\chi^2 = 19.87$, respectively). Untabulated test results indicate that the coefficient on $Resignation_{B4}$ is significantly larger than that on $Resignation_{NB4}$ (i.e., $\beta_1 > \beta_2$.). Thus, Big 4 resignations lead to more internal control problems than non-Big 4 resignations. In Model 3, we include four indicator variables that incorporate the size of the predecessor and successor auditors. We find that the four resignation indicator variables are statistically significant at the 5 percent level.

Because our regression model controls for the pre-switching information environment and future delisting or future incidence of lawsuits, *Resignation* is likely to capture auditors' proprietary information about clients' hidden future audit risks. Our results are consistent with auditors resigning from an engagement when they deem the client as being excessively high audit risk because of internal control issues, which is incrementally important to the pre-resignation information environment, to any future incidence of lawsuits, or to future delisting. Our results suggest that audit risk concerns are even more important when the predecessor auditor is a Big 4 firm.

### *Post-Resignation Delisting*

Table 6 presents the results of the relationship between auditor resignations and clients being delisted from the stock exchange in the post-resignation period because of (1) liquidation, (2) bankruptcy, (3) failure to meet minimum price, number of shareholders, financial or asset requirements, or (4) a corporate governance violation (our proxy for the predecessor auditors' exposure to future business risk had they not resigned from an engagement). The univariate results from Panel A of Table 6 indicate that the likelihood of a future stock exchange delisting is 10.83 percent for the *Resignation* sample and 6.62 percent for the *Dismissal* sample. The difference in the frequency of future lawsuits between the two samples (4.21 percent) is highly statistically significant (p-value $< 0.01$). This difference is even larger when a Big 4 auditor (5.12 percent) resigns from an engagement. Our results suggest that a Big 4 auditor resignation increases the odds

## TABLE 5

## Auditor Resignations and Future Internal Control Problems

### Panel A: Frequency of Internal Control Problems

| | *Resignation* | | *Dismissal* | | Differences | |
|---|---|---|---|---|---|---|
| | Number of Obs. | Mean | Number of Obs. | Mean | Mean | t-value |
| *Resignation* | 1,034 | 0.4526 | 4,410 | 0.2964 | 0.1562 | 9.22*** |
| *Resignation*$_{B4}$ | 494 | 0.4676 | 4,410 | 0.2964 | 0.1712 | 7.29*** |
| *Resignation*$_{NB4}$ | 540 | 0.4389 | 4,410 | 0.2964 | 0.1425 | 6.35*** |

### Panel B: Logistic Regressions

| | Model 1 | | Model 2 | | Model 3 | |
|---|---|---|---|---|---|---|
| Intercept | −0.6911 | (29.68)*** | −0.6856 | (28.94)*** | −0.6857 | (28.93)*** |
| *Resignation* | 0.2992 | (39.04)*** | | | | |
| *Resignation*$_{B4}$ | | | 0.3188 | (24.07)*** | | |
| *Resignation*$_{NB4}$ | | | 0.2811 | (19.87)*** | | |
| *Resignation*$_{B4toB4}$ | | | | | 0.3295 | (7.41)*** |
| *Resignation*$_{B4toNB4}$ | | | | | 0.3148 | (17.97)*** |
| *Resignation*$_{NB4toB4}$ | | | | | 0.4685 | (4.86)** |
| *Resignation*$_{NB4toNB4}$ | | | | | 0.2655 | (16.53)*** |
| Control Variables | | | | | | |
| *Acquisition* | 0.0558 | (0.91) | 0.0558 | (0.91) | 0.0560 | (0.92) |
| *Leverage* | −0.0082 | (0.10) | −0.0079 | (0.10) | 0.0078 | (0.10) |
| *Size* | 0.0089 | (0.65) | 0.0081 | (0.53) | 0.0076 | (0.46) |
| *Growth* | 0.0423 | (5.63)** | 0.0422 | (5.62)** | 0.0425 | (5.71)** |
| *Market-to-Book* | 0.0012 | (0.34) | 0.0012 | (0.35) | 0.0012 | (0.36) |
| *Return-on-Assets* | −0.0172 | (0.73) | −0.0172 | (0.72) | −0.0173 | (0.73) |
| *Litigation risk score* | 0.3339 | (110.12)*** | 0.3325 | (108.17)*** | 0.3327 | (108.27)*** |
| *Audit risk score* | 0.1274 | (28.37)*** | 0.1272 | (28.27)*** | 0.1268 | (28.04)*** |
| *Business risk score* | −0.0118 | (0.23) | −0.0120 | (0.23) | −0.0113 | (0.20) |
| *Lawsuit* | 0.5337 | (42.13)*** | 0.5336 | (42.08)*** | 0.5334 | (42.01)*** |
| *Delisting* | 0.0521 | (0.53) | 0.0517 | (0.53) | 0.0522 | (0.54) |
| Observations | 5,444 | | 5,444 | | 5,444 | |
| Adjusted $R^2$ (%) | 0.1833 | | 0.1834 | | 0.1836 | |
| Likelihood Ratio | 767.39*** | | 767.59*** | | 768.43*** | |
| Industry/Year fixed effects | Included | | Included | | Included | |

***, ** Indicate significance at the 1 percent and 5 percent levels, respectively.
The dependent variable *Internal Control* is 1 when a firm reports internal control problems within three years subsequent to an auditor switch. Subscript *B4* (*NB4*) denotes a Big 4 (non-Big 4) predecessor auditor. Subscript *B4toB4* (*NB4toNB4*) denotes a Big 4 (non-Big 4) predecessor and successor auditors. Subscript *B4toNB4* (*NB4toB4*) denotes a Big 4 (non-Big 4) predecessor auditor and a non-Big 4 (Big 4) successor auditor.

Variable Definitions:
*Resignation* = an indicator variable that equals 1 when an auditor resigns, and 0 otherwise;
*Acquisition* = an indicator variable that equals 1 for merger or acquisition, and 0 otherwise;
*Leverage* = ratio of the sum of long-term debt and short-term debt to total assets;
*Size* = logarithm of the book value of total assets;
*Growth* = percentage change in revenues between the current and the prior year;
*Market-to-Book* = ratio of the sum of the market value of equity plus the book values of preferred stock and debt divided by the book value of total assets;

*(continued on next page)*

**TABLE 5 (continued)**

*Return-on-Assets* = income before extraordinary items to total assets;
*Litigation risk score* = an *ex ante* metric for litigation risk based on the four variables defined in Table 2;
*Audit risk score* = an *ex ante* metric for audit risk based on the five variables defined in Table 2;
*Business risk score* = an *ex ante* metric for business risk based on the three variables defined in Table 2;
*Lawsuit* = an indicator variable that equals 1 when a firm is involved in a lawsuit within three years subsequent to an auditor switch; and
*Delisting* = an indicator variable that equals 1 when a firm is delisted from a national stock exchange within three years subsequent to an auditor switch.

of a future delisting by about 77 percent. In contrast, the likelihood of a future delisting is the smallest when a non-Big 4 auditor resigns.

In Table 6, Panel B we estimate Equation (2) using a logistic regression where the bivariate dependent variable *Delisted* is equal to 1 when a firm is delisted from the stock exchange within three years subsequent to an auditor switch, and 0 otherwise. In Model (1), the coefficient on *Resignation* ($\beta_1$) is positive and significant (0.3185, $\chi^2 = 22.87$; p-value $< 0.01$). In Model 2, we replace *Resignation* with $Resignation_{B4}$ and $Resignation_{NB4}$, the coefficients on $Resignation_{B4}$ ($\beta_1$) and $Resignation_{NB4}$ ($\beta_2$) are both positive and significant (0.3640, $\chi^2 = 17.58$ and 0.2706, $\chi^2 = 9.12$, respectively). Untabulated test results indicate that the coefficient on $Resignation_{B4}$ is significantly larger than that on $Resignation_{NB4}$ (i.e., $\beta_1 > \beta_2$.). Thus, while Big 4 and non-Big 4 auditor resignations lead to more frequent future delistings, the probability of a future delisting is even higher when a Big 4 auditor resigns from an engagement. In Model 3, we include four indicator variables that incorporate the size of the predecessor and successor auditors. We find that of the four *Resignation* indicator variables, only the coefficients on $Resignation_{B4toNB4}$ and $Resignation_{NB4toNB4}$ are statistically significant, the other two indicator variables are insignificant.

Because our regression model controls for the pre-switching information environment and future internal control problems and future incidence of lawsuits, *Resignation* is likely to capture auditors' proprietary information about clients' hidden future business risks, which becomes public information over the post-switching period when a client gets delisted. Our results are consistent with auditors resigning from an engagement when they deem the client as being unprofitable because of high business risk, which is incrementally important to the pre-resignation information environment, any future incidence of lawsuits, or to any future internal control problems. Our results suggest that business risk concerns are even more important when the predecessor auditor is a Big 4 firm.

## Sensitivity Analysis

### *Matched Sample*

We use a *Dismissal* sample as comparative group for the *Resignation* sample. However, there might be differences in the two subsamples that we fail to capture in our analyses. Therefore, we also construct a matched sample to analyze the *Resignation* sample, as follows. For each resignation observation, we find a matched dismissal observation that has the closest litigation risk value based on Shu's (2000) measure. Because we match without replacement, our procedure matches a unique dismissal observation to each resignation observation. Using this matching procedure, we are able to find 512 uniquely matched pairs.

When we use this matched sample as our comparative group for the *Resignation* sample, we get consistent results. For instance, in our *ex ante* tests, we find that *Business risk score* and *Audit risk score* are both positive and highly significant (0.4267, $\chi^2 = 24.62$ and 0.3094, $\chi^2 = 17.40$, respectively). These coefficients are much larger in magnitude than those in Table 3, which

## TABLE 6
## Auditor Resignations and Future Delisting

### Panel A: Frequency of Delisting

| | *Resignation* | | *Dismissal* | | Differences | |
|---|---|---|---|---|---|---|
| | **Obs.** | **Mean** | **Obs.** | **Mean** | **Mean** | **t-value** |
| *+/− Resignation* | 1,034 | 0.1083 | 4,410 | 0.0662 | 0.0421 | 4.06*** |
| $Resignation_{B4}$ | 494 | 0.1174 | 4,410 | 0.0662 | 0.0512 | 3.42*** |
| $Resignation_{NB4}$ | 540 | 0.1000 | 4,410 | 0.0662 | 0.0338 | 2.51*** |

### Panel B: Logistic Regressions

| | Model 1 | | Model 2 | | Model 3 | |
|---|---|---|---|---|---|---|
| Intercept | −2.1342 | (74.70)*** | −2.1131 | (72.57)*** | −2.1170 | (72.75)*** |
| *Resignation* | 0.3185 | (22.87)*** | | | | |
| $Resignation_{B4}$ | | | 0.3640 | (17.58)*** | | |
| $Resignation_{NB4}$ | | | 0.2706 | (9.12)*** | | |
| $Resignation_{B4toB4}$ | | | | | 0.2724 | (2.96) |
| $Resignation_{B4toNB4}$ | | | | | 0.3982 | (16.23)*** |
| $Resignation_{NB4toB4}$ | | | | | 0.1853 | (0.34) |
| $Resignation_{NB4toNB4}$ | | | | | 0.2795 | (9.14)*** |
| Control Variables | | | | | | |
| *Acquisition* | −0.1005 | (1.36) | −0.1006 | (1.37) | −0.1018 | (1.40) |
| *Leverage* | −0.2905 | (9.84)*** | −0.2853 | (9.47)*** | −0.2849 | (9.43)*** |
| *Size* | −0.0194 | (1.41) | −0.0223 | (1.78) | −0.0211 | (1.59) |
| *Growth* | 0.0049 | (0.03) | 0.0046 | (0.03) | 0.0043 | (0.02) |
| *Market-to-Book* | −0.0262 | (6.38)** | −0.0263 | (6.45)** | −0.0263 | (6.45)** |
| *Return-on-Assets* | 0.0661 | (1.58) | 0.0664 | (1.58) | 0.0661 | (1.56) |
| *Litigation risk score* | 0.1114 | (6.29)** | 0.1079 | (5.83)** | 0.1070 | (5.73)** |
| *Audit risk score* | 0.0763 | (4.94)** | 0.0760 | (4.90)** | 0.0758 | (4.88)** |
| *Business risk score* | 0.0638 | (3.00)* | 0.0621 | (2.82)* | 0.0613 | (2.75)* |
| *Lawsuit* | 0.4190 | (16.41)*** | 0.4192 | (16.41)*** | 0.4209 | (16.53)*** |
| *Internal Control* | 0.0573 | (0.90) | 0.0569 | (0.88) | 0.0578 | (0.91) |
| Observations | 5,444 | | 5,444 | | 5,444 | |
| Adjusted $R^2$ (%) | 0.1022 | | 0.1025 | | 0.1027 | |
| Likelihood Ratio | 233.46*** | | 234.11*** | | 234.70*** | |
| Industry/Year fixed effects | Included | | Included | | Included | |

***, **, * Indicate significance at the 1 percent, 5 percent, and 10 percent levels, respectively.
The dependent variable *Delisting* is 1 when a firm is delisted from a national stock exchange within three years subsequent to an auditor switch. Subscript *B4* (*NB4*) denotes a Big 4 (non-Big 4) predecessor auditor. Subscript *B4toB4* (*NB4toNB4*) denotes a Big 4 (non-Big 4) predecessor and successor auditors. Subscript *B4toNB4* (*NB4toB4*) denotes a Big 4 (non-Big 4) predecessor auditor and a non-Big 4 (Big 4) successor auditor.

Variable Definitions:
*Resignation* = an indicator variable that equals 1 when an auditor resigns, and 0 otherwise;
*Acquisition* = an indicator variable that equals 1 for merger or acquisition, and 0 otherwise;
*Leverage* = ratio of the sum of long-term debt and short-term debt to total assets;
*Size* = logarithm of the book value of total assets;
*Growth* = percentage change in revenues between the current and the prior year;
*Market-to-Book* = ratio of the sum of the market value of equity plus the book values of preferred stock and debt divided by the book value of total assets;

*(continued on next page)*

**TABLE 6 (continued)**

*Return-on-Assets* = income before extraordinary items to total assets;
*Litigation risk score* = an *ex ante* metric for litigation risk based on the four variables defined in Table 2;
*Audit risk score* = an *ex ante* metric for audit risk based on the five variables defined in Table 2;
*Business risk score* = an *ex ante* metric for business risk based on the three variables defined in Table 2;
*Lawsuit* = an indicator variable that equals 1 when a firm is involved in a lawsuit within three years subsequent to an auditor switch; and
*Internal Control* = an indicator variable that equals 1 when a firm reports internal control problems within three years subsequent to an auditor switch.

increases our confidence in our conclusion that business risk and audit risk are incrementally important considerations in resignation decisions. The coefficient on *Litigation risk score* is insignificant, which is expected because the sample is matched using Shu's (2000) litigation risk measure, which fully incorporates any difference in litigation risk between the resignation and dismissal subsamples and therefore allows us to focus on potential differences in business and audit risks.

### *Firm Fixed Effects*

One concern in cross-sectional studies is that the association between the variables of interest and the dependent variable could be attributable to correlated omitted variables. Himmelberg, Hubbard, and Palia (1999) argue that the inclusion of firm fixed effects largely addresses the concern because this form of specification controls for all firm-specific omitted variables. Therefore, we replicate all the results after including firm fixed effects in each specification. We find that the reported results are not affected when we include firm fixed effects.

### *Arthur Andersen and Second-Tier Auditors*

When we exclude Arthur Andersen's clients switching auditors in 2002 to eliminate client-auditor realignments from the demise of Andersen, the empirical results are similar to those reported. When we also exclude second-tier auditors, such as Grant Thornton and BDO Seidman, among the non-Big 4 due to the increasing visibility of the second-tier auditors, the results and conclusions remain unchanged.

## CONCLUSIONS

Our objective in this study is to examine the incremental importance of litigation risk, audit risk, and business risk as competing risk-based explanations for auditor resignation decisions using data from the pre- and post-resignation periods. We construct *ex ante* summary risk scores using public information one year prior to the resignation and then examine their relative and incremental importance. More important, we provide added evidence on the risk factors using client-related adverse outcomes from the post-resignation period. Drawing on the Bockus and Gigler (1998) model, we examine whether resignations lead to (1) more frequent future shareholder class-action lawsuits (future litigation risk proxy), (2) clients reporting problems with internal control over financial reporting (future audit risk proxy), and (3) more frequent delisting of the client from the stock exchange (future business risk proxy). In our post-resignation analyses, we include the *ex ante* risk indices and other control variables from the pre-resignation period, which allows us to isolate auditors' proprietary information about the clients' hidden risks that might have precipitated the resignation.

Based on a comprehensive sample of auditor switches between 1999 and 2010, we find that all three *ex ante* risk factors are incrementally important in explaining auditor resignations. Although audit and business risk are important determinants, litigation risk is the most important factor for Big 4 auditor resignations. The evidence from the post-switching period provides corroborating results.

Our study adds to our understanding of the importance of the different risk factors in auditor resignations. Our results support the argument that all three risk factors, and not just litigation risk, are important in resignation decisions. Finally, by analyzing the association between resignations and post-resignation outcomes while controlling for the pre-resignation public information, we are able to focus on the auditors' private information about the type of hidden risks that might have led to the resignation decisions.

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